

Chapter 6-8 cvd pvd 外延

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1. In the vacuum system, metal atoms get enough energy and get out of the metal surface, and deposit on the wafer, the atoms are non-directional, so the step coverage is worse than sputtering methods. While with sputtering method, the ions with electric field speed up, the ions are directional, so they have better step coverage.

2. according to the groove-model, the chemical reaction ratio are related to two factors.

when

$$h_g \gg k_s, C_s - > C_g \text{ reaction control}$$

$$h_g \ll k_s, C_s - > 0 \text{ diffusion control}$$

when the gas velocity is relative small, the reaction speed depends on diffusion

$$h_g \ll k_s$$

when velocity is big enough, it turns to reaction control

$$G - > \text{Saturate}$$

If we want to further enhance the deposition, we can have a higher temperature to have a faster reaction speed.

3. For homoepitaxy we have si-on-si and GaAs-on-GaAs. For heteroepitaxy we have Si-on-SiGe and SiGe-on-Si.

Lattice structure: In homoepitaxy, the lattice structure of the epitaxial layer is the same as that of the substrate, so there is no lattice matching problem. In heteroepitaxy, the lattice structure of the epitaxial layer is different from that of the substrate, so it is necessary to consider the problem of lattice matching, which may introduce strain.

Band structure: The band structure of homoepitaxy is similar to that of the substrate because they have the same chemical composition. However, in heteroepitaxy, due to the different chemical compositions of epitaxial layer and substrate, the energy band structure may change, which may affect the electrical properties of the material. **Electrical properties:** Homoepitaxial materials usually have better electrical properties because their lattice structure and chemical composition are the same as those of the substrate, so there are fewer defects and impurities at the interface. However, heteroepitaxy may introduce interfacial defects and strain, thus affecting the electrical properties of materials.